RAJALAKSHMI ENGINEERING COLLEGE

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AI RESUME ANALYZER APP

Submitted by

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BONAFIDE CERTIFICATE

Certified that this Project titled "AI Resume Analyzer App" is the bonafide work of "DHARANEEISH B K (2116230701072)" who carried out the work under my supervision. Certified further that to the best of my knowledge, the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

Submitted for the End semester practical examination to be held on ______________________

INTERNAL EXAMINER ***EXTERNAL EXAMINER***

ABSTRACT

In today's competitive digital environment, creating a professional and ATS-friendly resume has become essential for job seekers. "AI Resume Analyzer App" is an intelligent application developed to analyze resumes automatically using Artificial Intelligence and Natural Language Processing techniques. The system is designed to help users improve their resumes by evaluating content quality, identifying missing skills, and providing personalized suggestions for enhancement. The application supports resume uploads in formats such as PDF and DOCX and processes them efficiently through an AI-powered backend system.

The application integrates several important functionalities including Resume Parsing, Skill Extraction, ATS Score Generation, Keyword Analysis, Resume Classification, and Recommendation Generation. These features assist users in understanding resume strengths and weaknesses while improving compatibility with Applicant Tracking Systems used by recruiters. The user interface is designed with a simple and interactive layout to ensure smooth navigation and user-friendly interaction.

This project demonstrates the practical implementation of AI-based resume evaluation systems and highlights how intelligent automation can simplify recruitment preparation and improve career opportunities for users.

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CHAPTER 1
INTRODUCTION

1.1 GENERAL

Each module in the AI Resume Analyzer App is designed to operate efficiently while working together to provide a complete resume evaluation system. The application allows users to upload resumes, extract important information, analyze technical and soft skills, calculate ATS compatibility scores, and receive intelligent suggestions for improvement. The system processes resume content using AI and Natural Language Processing techniques to identify missing keywords, formatting issues, and skill gaps.

The user interface is developed with a clean and modern design to ensure easy navigation and better accessibility for users across different devices. Interactive screens and simplified workflows help users quickly upload resumes and view detailed analysis reports without complexity. The application architecture is optimized for fast processing, reliable performance, and smooth communication between frontend and backend components. Overall, the system provides an efficient and user-friendly platform that supports smarter resume building and career preparation.

1.2 SCOPE OF THE WORK

This project focuses on the design and development of an AI Resume Analyzer App that automates resume evaluation using Artificial Intelligence and Natural Language Processing techniques. The scope of the project includes implementing modules such as Resume Upload, Resume Parsing, Skill Extraction, ATS Score Generation, Keyword Analysis, and Recommendation Generation. The application is developed using Flutter for the frontend and FastAPI with Python for backend processing. The system analyzes resumes efficiently and provides intelligent feedback to improve resume quality and recruitment readiness.

In addition, the project focuses on providing a user-friendly interface with responsive layouts and smooth navigation for efficient resume analysis. Special attention is given to fast processing, accurate skill extraction, and optimized backend performance to ensure reliable operation. Overall, the AI Resume Analyzer App is designed to help users improve resume quality and increase ATS compatibility efficiently.

1.3 AIM AND OBJECTIVES OF THE PROJECT

The primary aim of this project is to design and implement an AI Resume Analyzer App that automates resume evaluation using Artificial Intelligence and Natural Language Processing techniques. The application is developed to help job seekers improve resume quality, increase ATS compatibility, and receive intelligent suggestions without depending entirely on manual review processes. The core objective is to provide a unified platform capable of analyzing resumes, extracting skills, identifying missing keywords, and generating detailed feedback efficiently.

To achieve this aim, the project focuses on developing multiple functional modules including Resume Upload, Resume Parsing, Skill Extraction, ATS Score Generation, Keyword Matching, Resume Classification, and Recommendation Generation. Each module is designed to work independently while seamlessly interacting with others to provide complete resume analysis. For example, extracted skills are compared with job-related keywords to calculate ATS scores and generate personalized improvement suggestions. The application supports PDF and DOCX resume formats for flexible user interaction.

A user-friendly design approach is adopted to ensure smooth navigation and better accessibility across devices. The system architecture emphasizes fast processing, accurate analysis, and efficient frontend-backend communication. Additionally, the modular development structure supports future enhancements such as interview preparation assistance, job recommendation systems, and multilingual resume analysis features.

CHAPTER 2
LITERATURE SURVEY

Recent research on AI-based recruitment systems and resume analysis applications has contributed significantly to the development of intelligent resume evaluation platforms. An automated resume screening system using Natural Language Processing and machine learning techniques is presented in [1]. The system extracts candidate skills, education, and experience from resumes and compares them with job descriptions to improve recruitment efficiency. Experimental results showed improved accuracy in identifying suitable candidates. This study supports the Resume Parsing and Skill Extraction modules of the AI Resume Analyzer App.

A resume classification system using keyword matching and NLP algorithms is proposed in [2]. The application analyzes resume content and generates ATS compatibility scores based on predefined recruitment criteria. The study demonstrated improved resume filtering and reduced manual screening time. This research supports the ATS Score Generation module in the AI Resume Analyzer App.

An intelligent recommendation system for resume improvement is discussed in [3]. The system identifies missing technical skills, formatting issues, and keyword gaps using AI-based analysis. Personalized suggestions are generated to improve resume quality and job relevance. This supports the Recommendation Generation module of the project.

A cloud-based resume analysis platform using FastAPI and machine learning models is described in [4]. The application processes PDF and DOCX resumes efficiently and provides structured feedback through a user-friendly interface. This research guides the backend implementation of the AI Resume Analyzer App.

A machine learning-based career recommendation and resume evaluation system is developed in [5]. The study highlights the effectiveness of AI in improving recruitment preparation and ATS optimization. This supports the overall functionality and intelligent analysis features of the AI Resume Analyzer App.

CHAPTER 3
SYSTEM DESIGN

3.1 ARCHITECTURE DIAGRAM

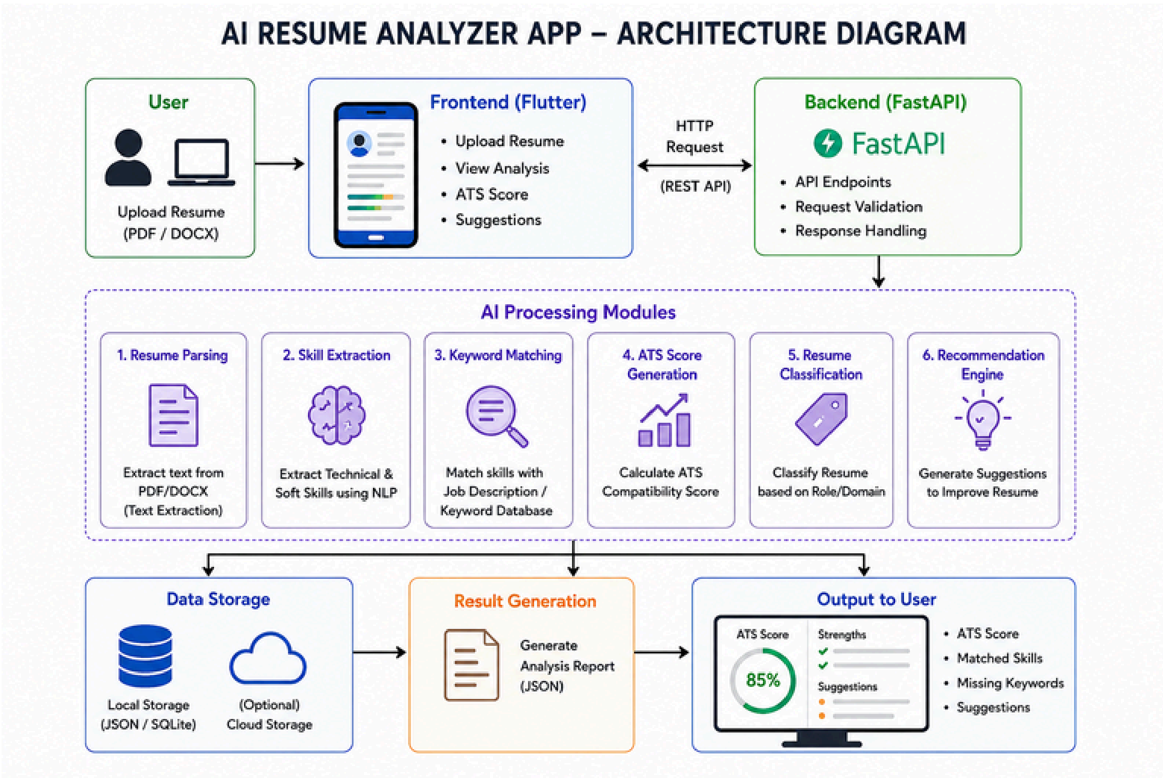

Fig 3.1.1 – Architecture Diagram

The AI Resume Analyzer App Architecture (refer Fig. 3.1.1) follows a modular and efficient design optimized for intelligent resume processing and smooth user interaction. Each core feature—Resume Upload, Resume Parsing, Skill Extraction, ATS Score Generation, Keyword Matching, Resume Classification, and Recommendation Engine—is developed as an independent module using Flutter, FastAPI, and Python-based AI techniques. Data flow across modules is handled efficiently to ensure accurate analysis and fast response generation.

Modules are managed through efficient frontend-backend communication and structured JSON responses, ensuring fast resume processing and reliable data handling. The system integrates AI and NLP techniques for resume parsing, skill extraction, and ATS score generation with minimal dependencies, keeping the application responsive and efficient. Modular UI components provide smooth navigation, clear result visualization, and seamless interaction between analysis features.

3.2 USE CASE DIAGRAM

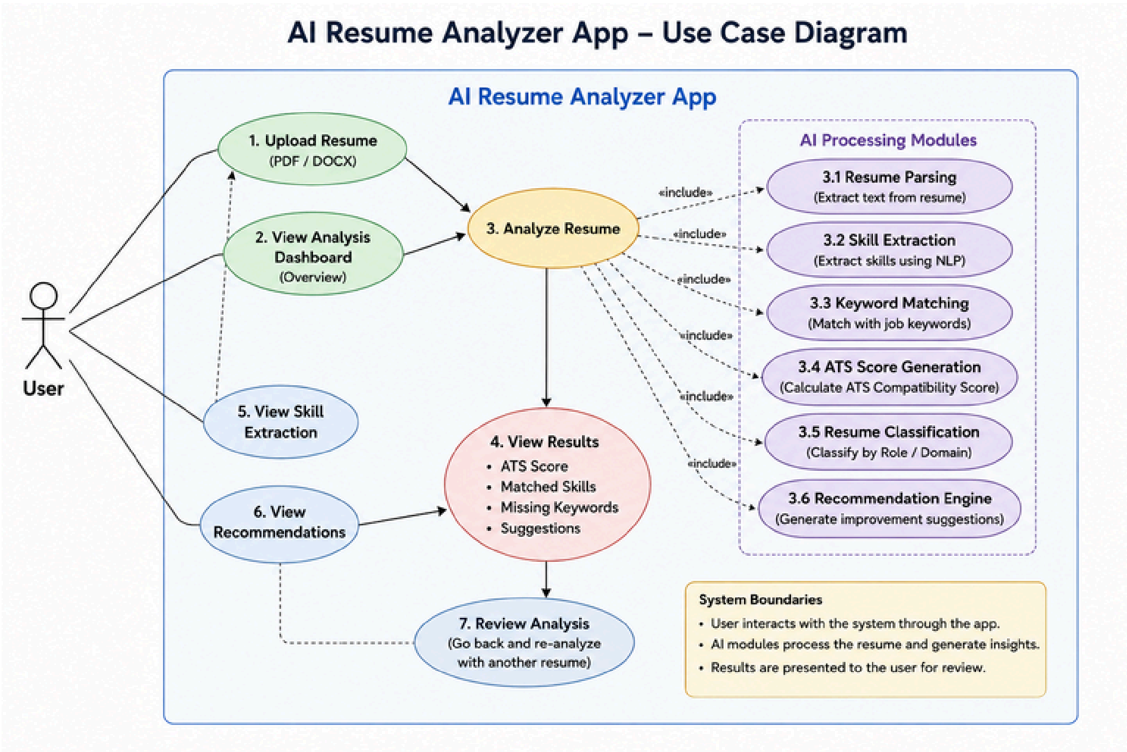

Fig 3.2.1 – Use Case Diagram

The Use Case Diagram for the AI Resume Analyzer App outlines the major functionalities designed to improve resume evaluation and career preparation. Users can upload resumes in PDF or DOCX format, analyze resumes using AI-based processing, and view ATS compatibility scores. The system performs Resume Parsing, Skill Extraction, Keyword Matching, and Resume Classification to generate accurate analysis results. The Recommendation Engine provides suggestions to improve resume quality, identify missing keywords, and enhance ATS performance for better recruitment opportunities.

These functions work together seamlessly to provide an intelligent platform for resume analysis, ATS evaluation, skill identification, and personalized recommendations, helping users improve resume quality and increase job opportunities efficiently.

3.3 HARDWARE SPECIFICATIONS

The AI Resume Analyzer App is designed to operate efficiently on standard smartphones and computing devices without requiring specialized hardware. The system architecture ensures smooth resume upload, AI-based analysis, ATS score generation, skill extraction, and recommendation processing on devices with moderate processing capability. The following hardware specifications are recommended for optimal performance:

Processor: Quad-core processor (e.g., Qualcomm Snapdragon 660 or higher, MediaTek Helio P60 or higher)

RAM: Minimum 3 GB (Recommended: 4 GB for smoother performance)

Storage: 32 GB internal storage or higher (to store user data and app files)

Graphics: Integrated graphics (no dedicated GPU required)

Operating System: Android 8.0 (Oreo) or higher

Other Requirements: Stable internet connection for cloud synchronization and data storage, touchscreen support, optional hardware for voice assistant features.

These specifications ensure the application runs efficiently, enabling smooth resume upload, fast AI-based analysis, accurate ATS score generation, and seamless user interaction across a variety of devices.

3.4 SOFTWARE SPECIFICATIONS

The AI Resume Analyzer App is designed using a modular software architecture to ensure efficient operation across various devices. The application integrates features such as Resume Upload, Resume Parsing, Skill Extraction, ATS Score Generation, Keyword Matching, and Recommendation Generation. The software stack is composed of the following technologies:

- ***Frontend Framework***:

  Flutter (for developing a responsive and user-friendly cross-platform interface with seamless interaction across mobile devices)

- ***Styling Tools***:

  Material Design (for clean and modern UI design)

- ***Backend Framework***:

  Natural Language Processing (NLP) Techniques (for resume parsing, skill extraction, keyword analysis, and intelligent ATS score generation without using a database system)

- ***Machine Learning Libraries***:

  Python Machine Learning Libraries (for implementing AI and NLP models used in resume analysis, skill extraction, keyword matching, and ATS score prediction)

- ***Database***:
  SharedPreferences and local JSON (for local data storage and quick access to user-specific data like tasks, notes, and settings)

- ***Data Handling & Preprocessing***:
  Python Standard Libraries and NLP Packages for efficient text processing, resume data extraction, and intelligent analysis operations.

CHAPTER 4
PROPOSED SYSTEM

The Proposed System for the AI Resume Analyzer App is designed to integrate intelligent resume evaluation features into one seamless platform, providing users with efficient ways to analyze and improve resumes. The system focuses on user-centric functionalities such as resume parsing, skill extraction, ATS score generation, keyword matching, and recommendation analysis.
Key components of the proposed system include:

Dashboard: Provides users with an overview of resume analysis results, ATS scores, matched skills, and recent activities in a single interface.

ATS Score Analysis: Evaluates resumes based on keyword matching, formatting, and job relevance to generate an ATS compatibility score.

Detailed Analysis: Performs resume parsing and skill extraction to identify strengths, missing keywords, technical skills, and improvement areas.

Improvement Suggestions: Generates personalized recommendations to enhance resume quality and increase job selection chances.

Job Matches: Suggests suitable job roles based on extracted skills, qualifications, and resume content analysis.

AI Chat Assistant: Assists users by answering resume-related queries, providing career guidance, and suggesting resume optimization tips.

Settings (Light/Dark Mode): Allows users to customize the application theme for better accessibility and user experience.

The system uses NLP-based processing and JSON responses for efficient resume analysis and data handling. The user interface is developed using Flutter, ensuring responsiveness, smooth navigation, and a user-friendly experience across different devices while maintaining fast AI-powered performance.

CHAPTER 5
MODULE DESCRIPTION

The AI Resume Analyzer App is designed with multiple intelligent modules to improve resume evaluation, ATS compatibility analysis, and career preparation. Below is an overview of each module:

1. Dashboard Module

The Dashboard provides users with a complete overview of resume analysis results, ATS scores, matched skills, and recent activities. It offers quick navigation to all major features and displays summarized analysis reports in a user-friendly interface.

2. ATS Score Analysis Module

This module evaluates resumes based on formatting, keyword matching, skills, and job relevance. It generates an ATS compatibility score that helps users understand how effectively their resume can pass Applicant Tracking Systems used by recruiters.

3. Detailed Analysis Module

The Detailed Analysis module performs resume parsing and extracts important information such as technical skills, soft skills, education, and experience. It identifies missing keywords, formatting issues, and areas requiring improvement using NLP techniques.

4. Improvement Suggestion Module

This module generates personalized recommendations to improve resume quality and increase job selection chances. Suggestions include adding missing skills, improving formatting, and optimizing keywords for better ATS performance.

5. Job Matches Module

The Job Matches module suggests suitable job roles based on extracted skills, qualifications, and resume analysis. It helps users identify career opportunities aligned with their profile.

6. AI Chat Assistant Module

The AI Chat Assistant helps users by answering resume-related queries, providing career guidance, and suggesting resume optimization strategies through interactive communication.

7. Settings Module

The Settings module allows users to customize application preferences such as Light Mode and Dark Mode for improved accessibility and user experience.

5.1 KEY BENEFITS OF PROPOSED SYSTEM

Enhanced Resume Analysis:

The application provides an efficient and intelligent way to analyze resumes, improving recruitment preparation by helping users identify strengths, missing skills, and ATS compatibility issues effectively.

AI-Based Skill Extraction:

With AI and NLP integration, users can automatically extract technical and soft skills from resumes, improving accuracy, accessibility, and resume evaluation speed.

ATS Score and Recommendations:

The ATS analysis module generates compatibility scores and personalized improvement suggestions, helping users optimize resumes for better job opportunities and recruiter visibility.

Easy-to-Use Dashboard:

The Dashboard feature provides a simple and organized interface for viewing analysis reports, matched skills, missing keywords, and recommendations, ensuring smooth navigation and better user experience.

CHAPTER-6

OUTPUT

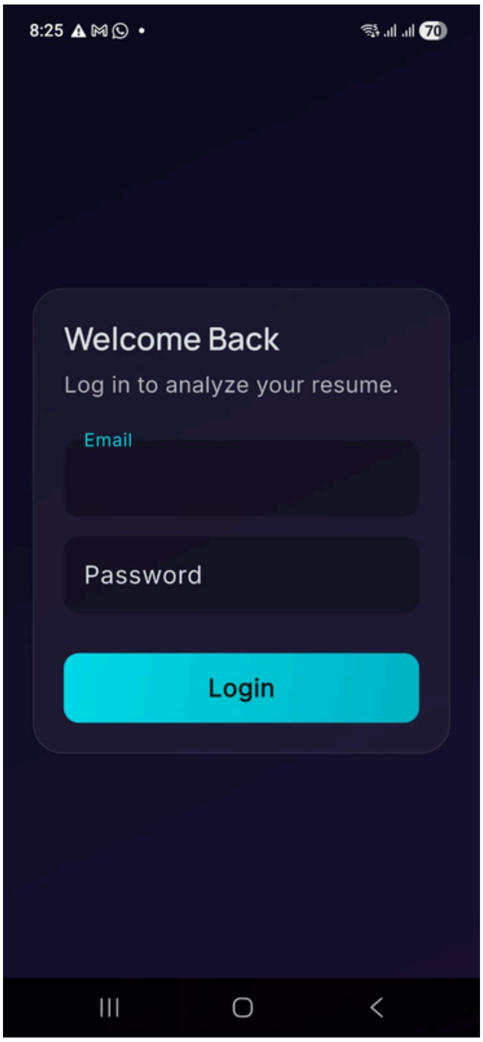

fig6.1 login page

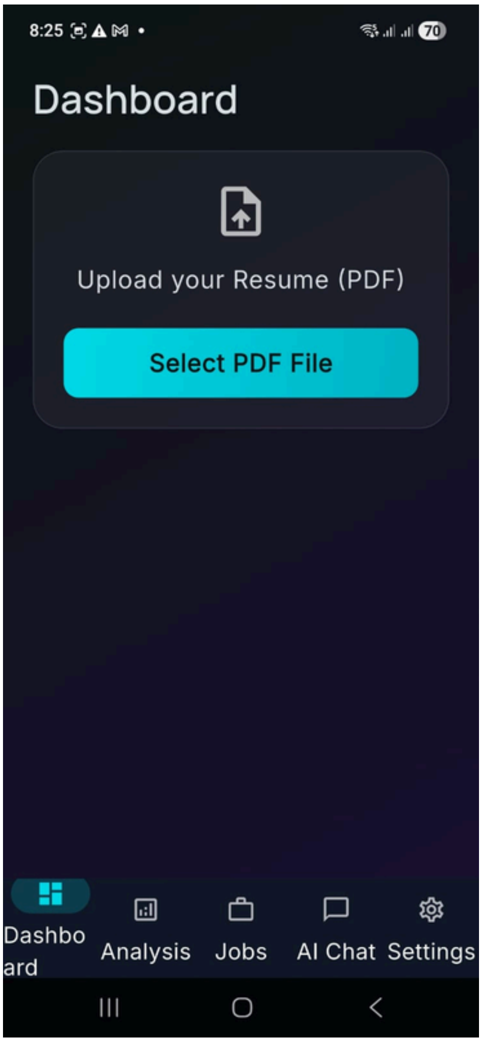

fig 6.2 Dashboard

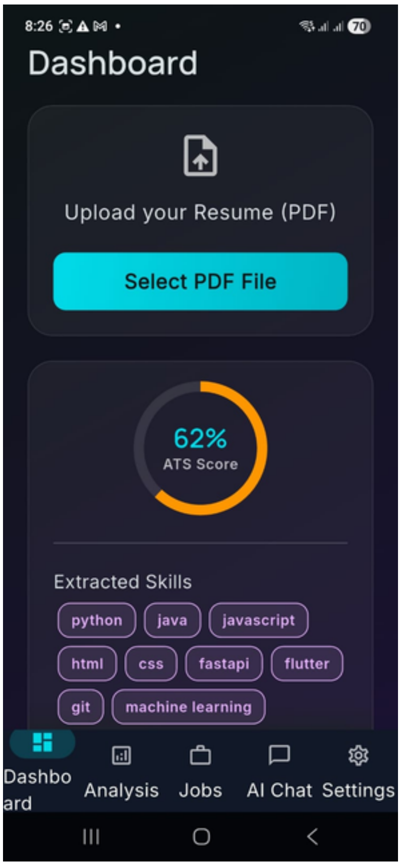

fig6.3 ATS Score

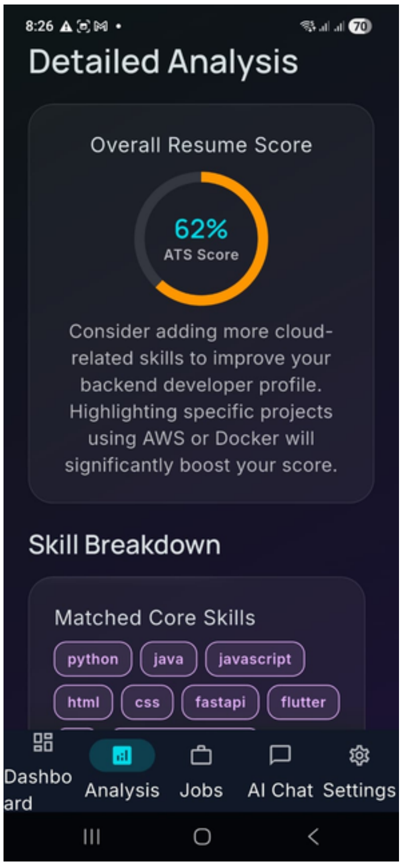

fig6.4 Detailed Analysis

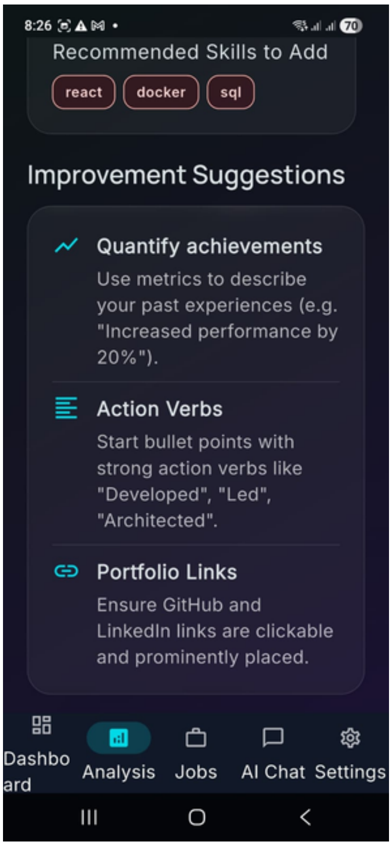

fig6.5 Improvement Suggestion

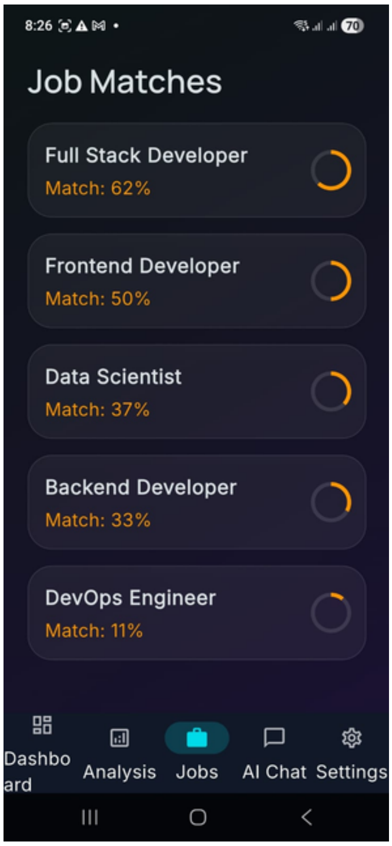

fig6.6 Job Matches

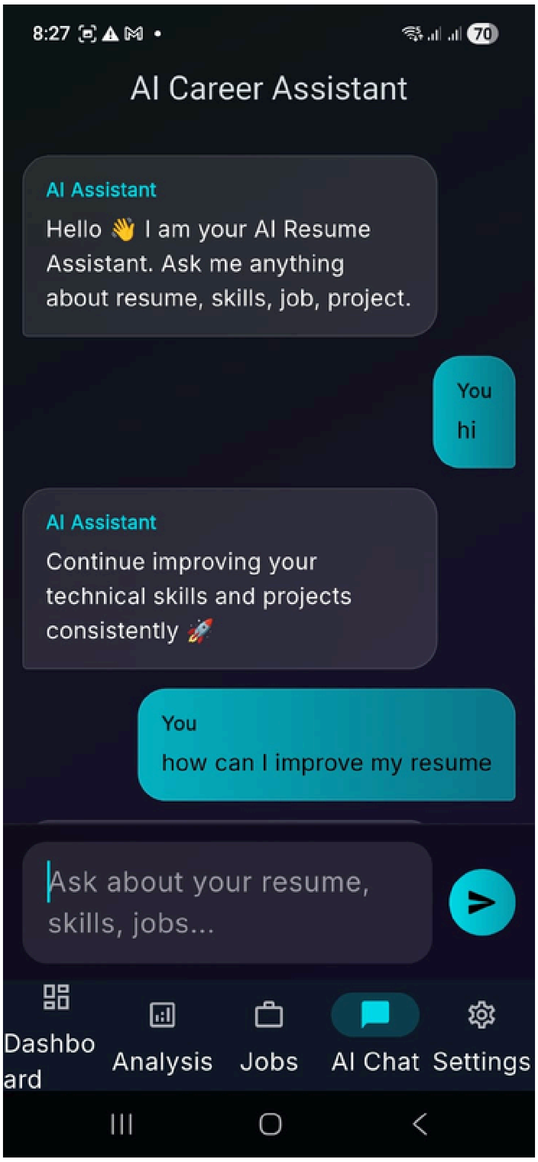

fig6.7 AI Chat Bot

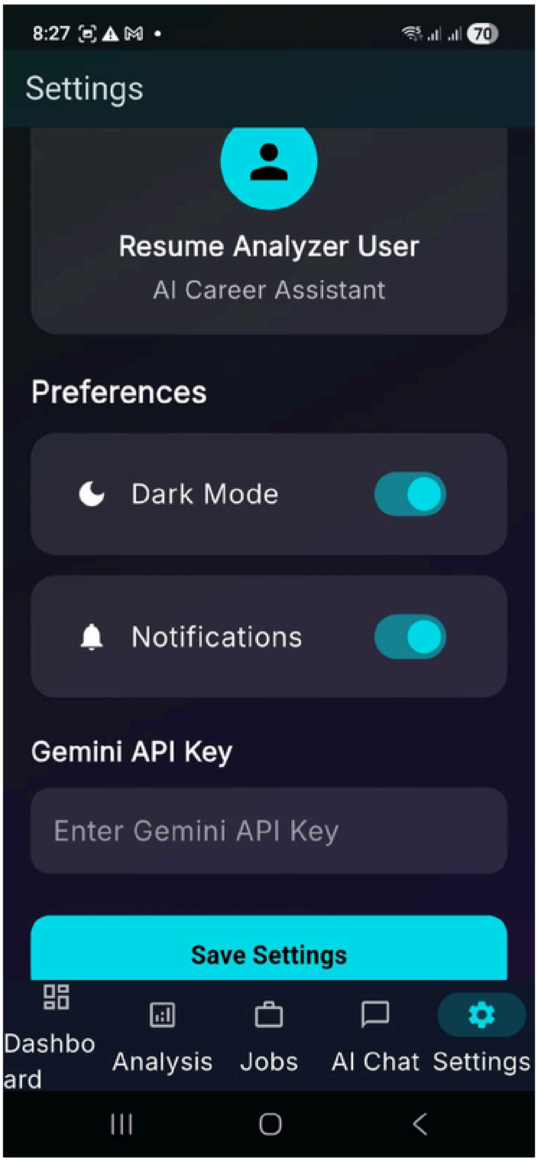

fig 6.8 Settings

CHAPTER 7

CONCLUSION AND FUTURE ENHANCEMENTS

The “AI Resume Analyzer App” provides users with an intelligent platform for analyzing resumes, improving ATS compatibility, and enhancing career opportunities. With features such as resume parsing, ATS score generation, skill extraction, detailed analysis, job matching, AI chat assistance, and personalized improvement suggestions, the application helps users create more effective resumes through a user-friendly interface. The integration of AI and NLP techniques ensures accurate analysis and efficient resume evaluation, enabling users to optimize resumes quickly and improve recruitment readiness anytime and anywhere.

Future Enhancements:

1. **Advanced AI Personalization:** Integrating more intelligent resume recommendations, personalized career guidance, and adaptive ATS optimization based on user profiles and resume analysis patterns.
2. **Cross-Platform Support:** Expanding the application to multiple platforms such as Android, iOS, and Web to provide a seamless resume analysis experience across devices.
3. **Integration with Job Portals:** Allowing integration with recruitment platforms and job portals such as LinkedIn, Indeed, and Naukri for direct job matching and application support.
4. **AI-Based Career Suggestions:** Implementing advanced AI-driven recommendations for career paths, skill development, and interview preparation based on resume content and industry trends.

CHAPTER 8
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